Structured questions

|!|BCELB0280000001|!| ★ **STSE** **Cross Topic**

CCEA GCE AL Life and Health Sciences Assessment Unit A2 5 Jun 2018 Q6

(a) Haemophilia is a sex-linked condition that affects the blood's ability to clot.
The gene for this condition is located on the X chromosome.
The allele (H) for normal blood clotting is dominant to the recessive allele (h) that causes haemophilia.

(i) Jessica is a carrier for haemophilia and has the genotype $X^H X^h$.
Jessica's husband Paul does not suffer from haemophilia. Give Paul's genotype. (1 mark)

(ii) Determine the probability of Jessica and Paul having a child who suffers from haemophilia. Show your working in a Punnett square. (4 marks)

(iii) If Jessica and Paul had another child, what would be the probability of this child having haemophilia? (1 mark)

(iv) Males are more likely to have haemophilia than females. Explain why. (2 marks)

© 2020 Aristo Educational Press Ltd.

(b) Genetic screening involves testing an individual for the presence of a genetic condition caused by a particular gene. Give one advantage and one disadvantage of genetic screening. (2 marks)

##

(a) (i) Paul: $X^H Y$ (1)

(ii)

		Male gametes	
		X^H	Y
Female gametes	X^H	$X^H X^H$	$X^H Y$
	X^h	$X^H X^h$	$X^h Y$

Correct gametes: X^H and X^h for Jessica (1)

X^H and Y for Paul (1)

Correct cross (1)

Probability is $^1/_4$ or 25% (1)

(iii) $^1/_4$ or 25% (1)

(iv) Males have only one X chromosome. (1)

If their X chromosome carries a recessive allele, they will have haemophilia. The chance is higher than females. (1)

OR

Females have two X chromosomes. (1)

They must inherit two recessive alleles to have haemophilia. The chance is smaller than males. (1)

(b) **Advantage**: (1)

Can make informed decisions / person knows if they are a carrier / know chance of having a child with genetic condition / may be less anxious

(Or other reasonable answers)

Disadvantage: (1)

Could decide not to have children / adverse effect on mental health / depression / expensive

(Or other reasonable answers)

(10 marks)

##

© 2020 Aristo Educational Press Ltd.

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Presenting information	3
	Mathematical skills	1
(a)(iii)	Mathematical skills	1
(a)(iv)	Applying concepts	2
(b)	Critical thinking	2

|!|BCELB0280000002|!| ★ **STSE** **Cross Topic**

If the concentration of glucose in the blood rises above normal, insulin secretion will be increased to bring the concentration back to normal.

(a) (i) Name the organ that secretes insulin. (1 mark)

_____

(ii) Describe the actions of insulin in bringing the concentration of glucose in the blood back to normal. (2 marks)

_____

_____

_____

_____

(b) Diabetics are unable to regulate their blood glucose concentrations naturally. In the 1980s, diabetics were treated with insulin extracted from pigs or cattle. With the advancement in biotechnology, human insulin can now be produced using recombinant DNA technology.

(i) Suggest why insulin has to be injected rather than taken orally. (2 marks)

_____

_____

_____

_____

© 2020 Aristo Educational Press Ltd.

(ii) What is recombinant DNA technology? (1 mark)

(iii) Suggest ***two*** advantages of producing human insulin using recombinant DNA technology over the traditional method of extracting insulin from pigs or cattle. (2 marks)

##

(a) (i) Pancreas (1)

(ii) Insulin stimulates liver cells to convert more glucose to glycogen for storage (1)

and stimulates body cells to take up more glucose from the blood for respiration. (1)

(b) (i) Insulin is a protein. (1)

When taken orally, insulin will be broken down by proteases in the digestive system (stomach). (1)

(ii) Recombinant DNA technology refers to the techniques involved in combining DNA molecules from different sources. (1)

(iii) Any ***two*** of the following: (2)

- The insulin produced has the same amino acid sequence as the insulin produced by the human body. / It has a lower chance of causing an immune response.
- The insulin produced is pure. / It is safer as the insulin extracted from animals may contain impurities.
- The production yield is higher. / It can be produced in a larger quantity in a shorter time.

(Accept other reasonable answers.)

(8 marks)

##

© 2020 Aristo Educational Press Ltd.

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	1
	Applying concepts	1
(b)(ii)	Applying concepts	1
(b)(iii)	Critical thinking	2

|!|BCELC0280000003|!| **Cross Topic**

Bacteria can be genetically modified to produce human proteins in large quantities. The steps involved are listed below. The order of the steps has been mixed up.

Step	Description
A	Mix recombinant plasmids with bacteria
B	Devise the nucleotide sequence by working backwards from the amino acid sequence
C	Identify successfully transformed bacteria
D	Isolate the desired protein from human cells and determine its amino acid sequence
E	Insert the artificial gene into cut plasmids
F	Harvest and purify the desired protein
G	Make the artificial gene by adding bases in sequence
H	Culture the transformed bacteria

(a) Arrange the above steps in the correct order. (2 marks)

D → ☐ → ☐ → ☐ → A → ☐ → ☐ → ☐

(b) Name the enzyme responsible for step E. (1 mark)

_____

(c) Suggest a reason why the artificial gene may have a different nucleotide sequence from the human gene. (1 mark)

_____

_____

© 2020 Aristo Educational Press Ltd.

(d) Describe how transformed bacteria produce the desired protein. (5 marks)

(e) Give ***one*** example of a protein that can be produced by the above method. (1 mark)

##

(a) D → B → G → E → A → C → H → F (2)

(b) DNA ligase (1)

(c) The genetic code is degenerate. / Most amino acids are coded by more than one code. (1)

(d) The artificial gene in the recombinant plasmid is transcribed to an mRNA molecule. (1)

The mRNA molecule binds to a ribosome. (1)

As the ribosome moves along the mRNA molecule, tRNA molecules carry amino acids to the ribosome. (1)

Amino acids are joined together by peptide bonds to form a polypeptide. (1)

When the ribosome reaches a stop codon on the mRNA, the polypeptide is released, and then coils and folds to form the desired protein. (1)

(e) Human insulin / Human growth hormone (1)

(Accept other reasonable answers.)

(10 marks)

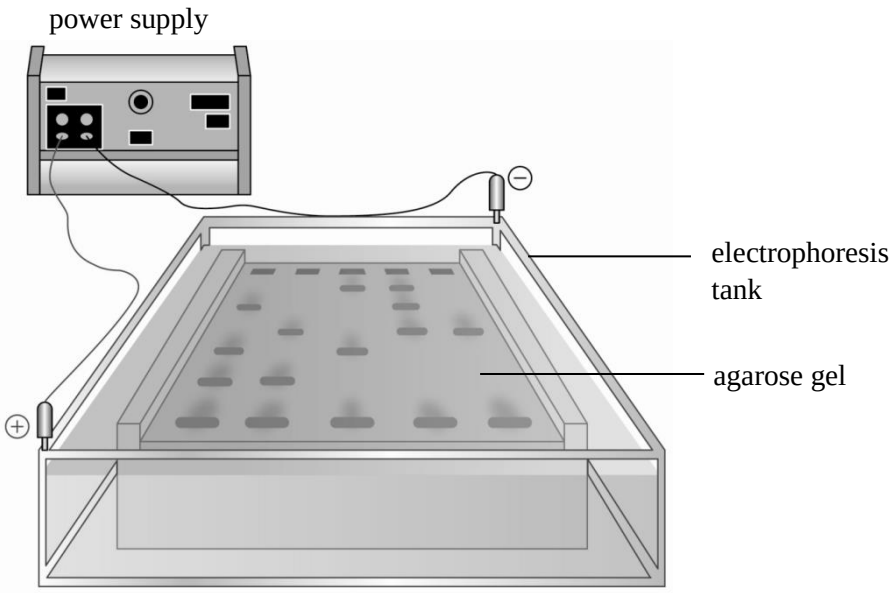
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Applying concepts	1
(d)	Understanding basic concepts	5
(e)	Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCELC0280000004|!|

The following diagram shows the set-up of gel electrophoresis for separating DNA fragments.



(a) At which end of the agarose gel (negative or positive) should DNA samples be loaded? Explain your answer. (3 marks)

(b) State ***two*** factors that may affect the speed at which the DNA fragments move through the gel. (2 marks)

(c) A solution is added into the electrophoresis tank. Name this solution and state its function. (2 marks)

© 2020 Aristo Educational Press Ltd.

(d) Describe how the DNA bands produced by gel electrophoresis can be observed. (1 mark)

##

(a) The DNA fragments should be loaded at the negative end of the agarose gel. (1)
This is because DNA is negatively charged (due to the presence of phosphate groups in DNA). (1)
When an electric current is applied to the gel, the DNA fragments will move towards the positive end. (1)

(b) Any ***two*** of the following: (2)
- voltage
- molecular size of the DNA fragments
- concentration of the agarose solution used to make the gel

(Accept other reasonable answers.)

(c) Buffer solution (1)
It maintains a constant pH value in the gel / allows an electric current to flow through the gel. (1)

(d) DNA bands can be observed by staining the gel with a dye (e.g. methylene blue).(1)

(8 marks)

##

	Skills assessed	Marks
(a)	Practical skills	3
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)	Practical skills	1

|!|BCELB0280000005|!| ★ **STSE** **Cross Topic**

The Human Genome Project (HGP) was an international research project that sequenced the human genome and determined the location of all human genes.

(a) Distinguish between a gene and a genome. (2 marks)

© 2020 Aristo Educational Press Ltd.

(b) State the term used to describe the position of a gene on a chromosome. (1 mark)

(c) Name ***one*** type of cell in the blood that can provide DNA samples for sequencing. (1 mark)

(d) Explain how the Human Genome Project has contributed to advances in medicine. (2 marks)

(e) State ***two*** drawbacks of the Human Genome Project. (2 marks)

##

(a) A gene is a DNA sequence that contains the information for synthesizing a particular protein. (1)
A genome is the complete set of DNA of an organism, including all the genes and other sequences. (1)

(b) Gene locus (1)

(c) White blood cells / Phagocytes / Lymphocytes (1)

(d) Any ***two*** of the following: (2)
- It improved the diagnosis of disorders by genetic testing.
- It helped the development of new treatments (e.g. gene therapy) for diseases.
- It helped personalize medication for patients.

(Accept other reasonable answers.)

© 2020 Aristo Educational Press Ltd.

(e) Any ***two*** of the following: (2)

- Disputes may arise about the ownership of the genetic information.
- The privacy of genetic information may be violated / Personal genetic information may be disclosed.
- It may lead to a negative public perception of / discrimination against people who have certain genetic differences.
- It may become easier to select specific genes to create 'designer babies'.

(Accept other reasonable answers.)

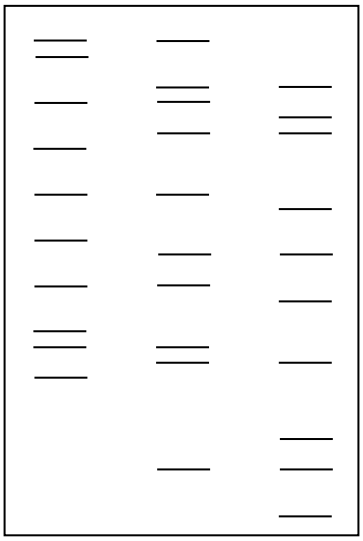
(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Applying concepts	1
(d)	Applying concepts	2
(e)	Applying concepts	2

|!|BCELB0280000006|!| ★

The diagram below shows the DNA fingerprints of persons X, Y and Z. These three persons are the parents and the son of a family.



(a) State ***one*** source of DNA that can be used in DNA fingerprinting. (1 mark)

© 2020 Aristo Educational Press Ltd.

(b) Outline the major steps of DNA fingerprinting. (3 marks)

(c) According to the DNA fingerprints shown above, deduce which of X, Y and Z are the parents and which is the son. (4 marks)

##

(a) Hairs / Blood / Saliva (or other reasonable answers) (1)

(b) Any ***three*** of the following: (3)

- DNA is cut into DNA fragments using restriction enzymes. / DNA fragments containing highly variable regions are obtained.
- The DNA fragments are loaded into an agarose gel slab and placed in an electric field.
- The DNA fragments are separated into bands according to their sizes by gel electrophoresis.
- The gel is stained with a dye to reveal the DNA bands.

(c) The son receives half of his genetic material from his father and receives the other half from his mother. (1)

From the DNA fingerprints, half of the DNA bands of person Y matches with those of person X while the other half matches with those of person Z. (1)

Therefore, persons X and Z are the parents (1)

and person Y is the son. (1)

(8 marks)

##

© 2020 Aristo Educational Press Ltd.

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)	Analysing data	1
	Applying concepts	3

© 2020 Aristo Educational Press Ltd.